Laboratorio 7:

1. Eliminar Epsilon:

$S \to 00 \mid 1B1 \mid BB \mid 0A0$

$S \to 0A0 \mid 11 \mid 1B1 \mid B$

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$

$A \to C$

$B \to S \mid A$

$C \to S$

2. Eliminar producciones unarios

$P_0 = \{(S,S), (A,A), (B,B), (C,C)\}$

(S,S)	$S \to B$	(S,B)	(A,A)	$A \to C$	(A,C)
(S,B)	$B \to A$	(S,A)	(A,C)	$C \to S$	(A,S)
(S,A)	$A \to C$	(S,C)			
(S,C)	$C \to S$	(S,S)	(C,C)	$C \to S$	(C,S)

(B,B)	$B \to A$	(B,A)
(B,A)	$A \to C$	(B,C)
(B,C)	$C \to S$	(B,S)

$S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$A \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$B \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$C \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

3. Eliminar inútiles

$T = \{0,1\}$

$\omega_1 = \{S,A,B,C\}$

$\omega_2 = \{S,A,B,C\}$

$Y_1 = 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$Y_2 = 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$S \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$A \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

$B \to 0A0 \mid 1B1 \mid 00 \mid 11 \mid BB$

4. Forma Chomsky:

$T_0 = 0 \quad T_1 = 1$

$S = T_0AT_0 \mid T_1BT_1 \mid 00 \mid 11 \mid BB$

$A = T_0AT_0 \mid T_1BT_1 \mid 00 \mid 11 \mid BB$

$B = T_0AT_0 \mid T_1BT_1 \mid 00 \mid 11 \mid BB$

$C_1 = AT_0 \quad C_2 = BT_1$

$S \to T_0C_1 \mid T_1C_2 \mid 00 \mid 11 \mid BB$

$B \to T_0C_1 \mid T_1C_2 \mid 00 \mid 11 \mid BB$

$A \to T_0C_1 \mid T_1C_2 \mid 00 \mid 11 \mid BB$

Ejercicio 2:

1. Eliminar ϵ:

$S \to aAa|aa|bBb$
$S \to aAa|bb|bBb$

$S \to aAa|aa|bb|bBb$
$A \to C|a$
$B \to C|b$
$C \to CDE|DE|CE|E$
$D \to A|B|ab$

2. Eliminar producciones unarios

$P_0 = \{(S,S),(A,A),(B,B),(C,C),(D,D)\}$

(C,C) $C \to E$ (C,E)

(S,S)

(A,A) $A \to C$ (A,C)
(A,C) $C \to E$ (A,E)

(B,B) $B \to C$ (B,C)
(B,C) $C \to E$ (B,E)

(D,D) $D \to A$ (D,A)
(D,A) $A \to C$ (D,C)
(D,C) $C \to E$ (D,E)
(D,D) $D \to B$ (D,B)
(D,B)

$(C,C): CDE|DE|CE$
$(C,E): —$
$(S,S): aAa|aa|bb|bBb$
$(A,A): a$
$(A,C): CDE|DE|CE$
$(B,B): b$
$(B,C): CDE|DE|CE$
$(D,D): ab$
$(D,A): a$
$(D,B): b$
$(D,C): CDE|DE|CE$

$S \to aAa|aa|bb|bBb$
$A \to CDE|DE|CE|a$
$B \to CDE|DE|CE|b$
$C \to CDE|DE|CE$
$D \to CDE|DE|CE|ab|a|b$

3. Eliminar inútiles o inalcanzables

$T : \{a, b$

$W_1 : \{S, A, B, D, a, b\}$

$W_2 : \{S, A, B, C, D, a, b\}$

$W_3 : \{S, A, B, C, D, a, b\}$

$Y_1 \rightarrow \{S\}$

$Y_2 \rightarrow \{D, A, B, a, b, S\}$

$Y_3 \rightarrow \{D, A, B, a, b, C, S\}$

$S \rightarrow aAb \mid aa \mid bb \mid bBb$

$A \rightarrow CDE \mid DE \mid CE \mid a$

$B \rightarrow CDE \mid DE \mid CE \mid b$

$C \rightarrow CDE \mid DE \mid CE$

$D \rightarrow CDE \mid DE \mid CE \mid ab \mid a \mid b$

4. Forma Chomsky

$C_1 = DE \quad T_1 = a \quad T_2 = b$

$S \rightarrow T_1AT_1 \mid T_1T_1 \mid T_2T_2 \mid T_2BT_2$

$A \rightarrow CC_1 \mid DE \mid CE \mid T_1$

$B \rightarrow CC_1 \mid DE \mid CE \mid T_2$

$C \rightarrow CC_1 \mid DE \mid CE$

$D \rightarrow CC_1 \mid DE \mid CE \mid T_1T_2 \mid T_1 \mid T_2$

$C_2 = AT_1 \quad C_3 = BT_2$

$S \rightarrow T_1C_2 \mid T_1T_1 \mid T_2T_2 \mid T_2C_3$

$A \rightarrow CC_1 \mid DE \mid CE \mid T_1$

$B \rightarrow CC_1 \mid DE \mid CE \mid T_2$

$C \rightarrow CC_1 \mid DE \mid CE$

$D \rightarrow CC_1 \mid DE \mid CE \mid T_1T_2 \mid T_1 \mid T_2$

Ejercicio 3:

1. Eliminar ϵ:

$S \to ASA \mid a \mid aB$

$S \to SA \mid AS \mid ASA \mid aB$

$S \to ASA \mid SA \mid AS \mid aB \mid a$

$S \to ASA \mid SA \mid AS \mid aB \mid a \mid S$

$A \to B \mid S$

$B \to b$

2. Eliminar p. unitarios

$P_0 = \{(S,S), (A,A), (B,B)\}$

(A,A) $A \to B$ (A,B)

(A,A) $A \to S$ (A,S)

(A,A) : —

(A,S) : $ASA \mid AS \mid SA \mid aB \mid a$

(A,B) : b

3. Eliminar inútiles

$T : \{a, b\}$

$W_1 : \{S, A, B, a, b\}$

$W_2 : \{S, A, B, a, b\}$

$Y_1 : \{S\}$

$Y_2 : \{S, a, A, b\}$

$Y_3 : \{S, A, B, a, b\}$

$Y_4 : \{S, A, B, a, b\}$

$S \to ASA \mid AS \mid SA \mid aB \mid a$

$A \to ASA \mid AS \mid SA \mid aB \mid a \mid b$

$B \to b$

4. Forma Chomsky

$C_1 = AS$ $T_1 = a$ $T_2 = b$

$S \to C_1A \mid AS \mid SA \mid T_1B \mid T_1$

$A \to C_1A \mid AS \mid SA \mid T_1B \mid T_1 \mid T_2$

$B \to T_2$